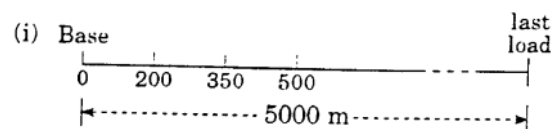


Q 1

2000

In the construction of a 5 km expressway a truck delivers materials from a base. After depositing each load, the truck returns to the base to collect the next load. The first load is deposited 200 m from the base, the second 350 m from the base, the third 500 m from the base. Each subsequent load is deposited 150 m from the previous one.

- How far is the fifteenth load deposited from the base?
- How many loads are deposited along the total length of the 5 km expressway? (The last load is deposited at the end of the expressway.)
- How many kilometres has the truck travelled in order to make all the deposits and then return to the base?



This is an arithmetic series, with $a = 200$,
 $d = 150$.

$$T_n = a + (n - 1)d$$

$$\therefore T_{15} = 200 + 14 \times 150$$

$$= 2300.$$

$\therefore$ The fifteenth load is deposited 2300 m from the base.

$$(ii) T_n = 5000,$$

$$5000 = 200 + (n - 1) \times 150$$

$$150(n - 1) = 4800$$

$$n - 1 = 32$$

$$n = 33.$$

$\therefore$ There are 33 loads.

$$(iii) \text{ Using } S_n = \frac{n}{2}(a + \ell),$$

total distance travelled

$$= 2 \times \left[\frac{33}{2}(200 + 5000) \right]$$

$$= 171600 \text{ m}$$

$$= 171.600 \text{ km.}$$

6

Q 2

2001

The first three terms of an arithmetic series are $-1 + 4 + 9 + \dots$

- Find the 60th term. 2
- Hence, or otherwise, find the sum of the first 60 terms of the series. 2

$$(a) a = -1, d = 5, n = 60.$$

$$(i) T_n = a + (n - 1)d$$

$$T_{60} = -1 + 59 \times 5$$

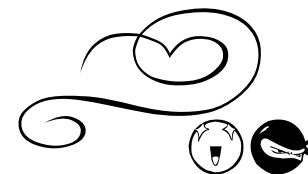
$$= 294.$$

$$(ii) S_n = \frac{n}{2}(a + \ell)$$

$$S_{60} = \frac{60}{2}(-1 + 294)$$

$$= 8790.$$



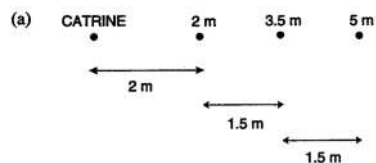


Q 3

2002

Catrine is exercising her dog by throwing a stick for the dog to fetch and return. The first time, Catrine throws the stick 2 m and she continues to throw the stick after the dog has returned it. Each time, she increases the distance the stick is thrown by exactly 1.5 m. Her last throw is 32 m.

- How many times did Catrine throw the stick?
- How far did her dog run altogether in fetching and returning the stick? (Assume that the dog starts and finishes at Catrine's side.)



(i) The throws constitute an arithmetic sequence:
2, 3.5, 5, ...

$$\begin{aligned}
 U_n &= a + (n-1)d \\
 U_n &= 32, \quad a = 2, \quad d = 1.5 \\
 \therefore 32 &= 2 + (n-1)1.5 \\
 32 &= 2 + 1.5n - 1.5 \\
 32 &= 0.5 + 1.5n \\
 31.5 &= 1.5n \\
 n &= \frac{31.5}{1.5} \\
 &= 21
 \end{aligned}$$

$\therefore$ Catrine threw the stick 21 times.

(ii) Outward distance run by the dog:

$$\begin{aligned}
 S_n &= \frac{n}{2} [2a + (n-1)d] \\
 a &= 2, \quad d = 1.5, \quad n = 21 \\
 S_n &= \frac{21}{2} [2(2) + (21-1)1.5] \\
 &= \frac{21}{2} [4 + (20)1.5] \\
 &= \frac{21}{2} [4 + 30] \\
 &= \frac{21}{2} \times 34 \\
 &= 357 \text{ m.}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Total distance run by dog} \\
 &= 2 \times 357 \text{ m} \\
 &= 714 \text{ m.}
 \end{aligned}$$

Q 4

2002

Consider the geometric series

$$1 + (\sqrt{5} - 2) + (\sqrt{5} - 2)^2 + \dots$$

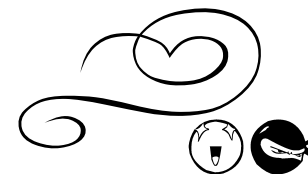
- Explain why the geometric series has a limiting sum. 1
- Find the exact value of the limiting sum. Write your answer with a rational denominator. 2

(i) A limiting sum exists if
 $|r| < 1$, i.e. $-1 < r < 1$

$$\begin{aligned}
 \text{Now } r &= \sqrt{5} - 2 \\
 &\approx 0.236 \quad (\text{by calc.}) \\
 \text{and } -1 &< 0.236 < 1
 \end{aligned}$$

$\therefore$ A limiting sum exists.

$$\begin{aligned}
 \text{(ii) } S &= \frac{a}{1-r} \\
 &= \frac{1}{1 - (\sqrt{5} - 2)} \\
 &= \frac{1}{1 - \sqrt{5} + 2} \\
 &= \frac{1}{3 - \sqrt{5}} \\
 &= \frac{1}{3 - \sqrt{5}} \times \frac{3 + \sqrt{5}}{3 + \sqrt{5}} \\
 &= \frac{3 + \sqrt{5}}{9 - 5} \\
 &= \frac{3 + \sqrt{5}}{4}
 \end{aligned}$$



Q 5

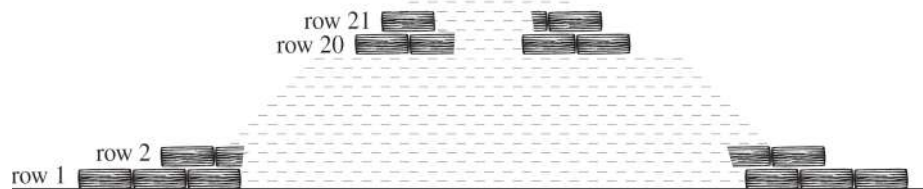
2003

A wall is built to stop erosion when a level road is cut through a hill.

The rows of the wall, built from concrete blocks 1.5 m long, are numbered from the bottom. The bottom row (row 1) is 180 m long.

Each of the rows 2, 3, ..., 20 has 3 fewer blocks than the row below it.

Above row 20, each row has 1 block fewer than the row below it. The top row has 10 blocks.



(i) How many blocks are in row 20?

2

(ii) What is the total number of rows in the wall?

1

(iii) How many blocks are used in the construction of the wall?

2

(i) Length of row 1 = 180 m
Length of each block = 1.5 m

$$\begin{aligned}\therefore \text{Number of blocks in row 1} &= \frac{180}{1.5} \\ &= 120.\end{aligned}$$

The number of blocks in each row up to row 20 are in an arithmetic sequence:

120, 117, 114, ... (to row 20)
with $a = 120$ and $d = -3$.

$$\begin{aligned}\text{Now } U_n &= a + (n-1)d \\ \therefore U_{20} &= 120 + 19(-3) \\ &= 120 - 57 \\ &= 63\end{aligned}$$

i.e. There are 63 blocks in row 20.

(ii) Now row 21 has $63 - 1 = 62$ blocks in it.

The number of blocks in each row from row 21 to the top row are in an arithmetic sequence:
62, 61, ..., 10
with $a = 62$ and $d = -1$.

$$\text{Now } U_n = a + (n-1)d$$

$$\begin{aligned}\text{If } U_n &= 10, \\ 10 &= 62 + (n-1)(-1) \\ 10 &= 62 - n + 1 \\ 10 &= 63 - n \\ n &= 63 - 10 \\ &= 53\end{aligned}$$

$$\begin{aligned}\therefore \text{Total number of rows in the wall} &= 20 + 53 \\ &= 73.\end{aligned}$$

(iii) $S_n = \frac{n}{2}(a + l)$

For lower part of wall (rows 1–20):
 $n = 20$, $a = 120$ and $l = 63$

$$\begin{aligned}\therefore S_{20} &= \frac{20}{2}(120 + 63) \\ &= 10 \times 183 \\ &= 1830\end{aligned}$$

For upper part of wall (rows 21–73):
 $n = 53$, $a = 62$ and $l = 10$

$$\begin{aligned}\therefore S_{53} &= \frac{53}{2}(62 + 10) \\ &= \frac{53}{2} \times 72 \\ &= 1908\end{aligned}$$

$$\begin{aligned}\therefore \text{Total number of blocks used} &= 1830 + 1908 \\ &= 3738.\end{aligned}$$

Q 6

2003

(i) Find the limiting sum of the geometric series

$$2 + \frac{2}{\sqrt{2} + 1} + \frac{2}{(\sqrt{2} + 1)^2} + \dots$$

(ii) Explain why the geometric series

$$2 + \frac{2}{\sqrt{2} - 1} + \frac{2}{(\sqrt{2} - 1)^2} + \dots$$

does NOT have a limiting sum.

$$(i) \quad S = \frac{a}{1-r}$$

Now $a = 2$,

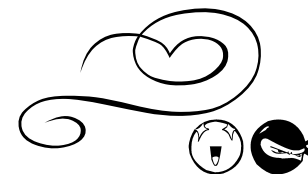
$$\text{and } r = \frac{2}{\sqrt{2} + 1}$$

$$\begin{aligned}&= \frac{1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} \\ &= \frac{\sqrt{2} - 1}{(\sqrt{2})^2 - 1^2} \\ &= \frac{\sqrt{2} - 1}{2 - 1} \\ &= \sqrt{2} - 1.\end{aligned}$$

$$\begin{aligned}\therefore S &= \frac{2}{1 - (\sqrt{2} - 1)} \\ &= \frac{2}{2 - \sqrt{2}} \times \frac{2 + \sqrt{2}}{2 + \sqrt{2}} \\ &= \frac{2(2 + \sqrt{2})}{2^2 - (\sqrt{2})^2} \\ &= \frac{2(2 + \sqrt{2})}{2} \\ &= 2 + \sqrt{2}.\end{aligned}$$

$$\begin{aligned}(ii) \quad r &= \frac{2}{\sqrt{2} - 1} \\ &= \frac{1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} \\ &= \frac{\sqrt{2} + 1}{(\sqrt{2})^2 - 1^2} \\ &= \frac{\sqrt{2} + 1}{2 - 1} \\ &= \frac{\sqrt{2} + 1}{1} \\ &= \sqrt{2} + 1 \\ &\neq 2.4\end{aligned}$$

For a limiting sum to exist, $-1 < r < 1$ and $\sqrt{2} + 1$ does not satisfy this inequality.



Q 7

2004

Consider the geometric series $1 - \tan^2 \theta + \tan^4 \theta - \dots$

(i) When the limiting sum exists, find its value in simplest form.

(ii) For what values of θ in the interval $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ does the limiting sum of the series exist?

(i) $1 - \tan^2 \theta + \tan^4 \theta - \dots$

This is a geometric series with $a = 1$,
 $r = -\tan^2 \theta$.

When the limiting sum exists,

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} \\ &= \frac{1}{1-(-\tan^2 \theta)} \\ &= \frac{1}{1+\tan^2 \theta} \\ &= \frac{1}{\sec^2 \theta} \\ &= \cos^2 \theta. \end{aligned}$$

ii) For the limiting sum to exist, $|r| < 1$.

ie. $-1 < r < 1$

$-1 < -\tan^2 \theta < 1$ (since $r = -\tan^2 \theta$)

$1 > \tan^2 \theta > -1$

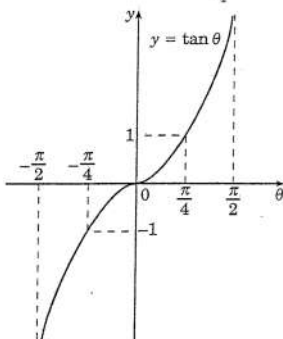
ie. $-1 < \tan^2 \theta < 1$.

Since $\tan^2 \theta > 0$, only need to solve
 $\tan^2 \theta < 1$

$\therefore -1 < \tan \theta < 1$.

When $\tan \theta = \pm 1$, $\theta = \pm \frac{\pi}{4}$.

Since $y = \tan \theta$ is an increasing function,
this means the solution is $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$.



Q 8

2004

Evaluate $\sum_{n=2}^4 n^2$.

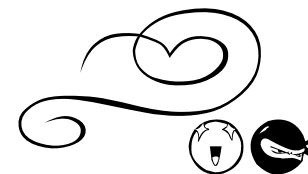
$$\begin{aligned} \sum_{n=2}^4 n^2 &= 2^2 + 3^2 + 4^2 \\ &= 4 + 9 + 16 \\ &= 29. \end{aligned}$$

Q 9

2005

Evaluate $\sum_{n=3}^5 (2n+1)$.

$$\begin{aligned} \sum_{n=3}^5 (2n+1) &= (2 \times 3 + 1) + (2 \times 4 + 1) + (2 \times 5 + 1) \\ &= 7 + 9 + 11 \\ &= 27. \end{aligned}$$



Q10

2004

Clare is learning to drive. Her first lesson is 30 minutes long. Her second lesson is 35 minutes long. Each subsequent lesson is 5 minutes longer than the lesson before.

- How long will Clare's twenty-first lesson be?
- How many hours of lessons will Clare have completed after her twenty-first lesson?
- During which lesson will Clare have completed a total of 50 hours of driving lessons?

(a) Lesson 1 = 30 minutes
Lesson 2 = 35 minutes
Lesson 3 = 40 minutes
∴ 30, 35, 40, 45, ...
This is an arithmetic sequence with $a = 30$, $d = 5$.

$$\begin{aligned} \text{(i)} \quad T_n &= a + (n-1)d \\ \therefore T_{21} &= 30 + (21-1) \times 5 \\ &= 30 + 20 \times 5 \\ &= 130. \\ \therefore \text{The 21st lesson will be 130 minutes} \\ &\text{long (ie. 2 h 10 min).} \end{aligned}$$

(ii) **METHOD 1**

$$\begin{aligned} S_n &= \frac{n}{2}(a + \ell) \\ a &= 30, \ell = 130, n = 21. \\ \therefore S_{21} &= \frac{21}{2}(30 + 130) \\ &= 1680 \text{ min} \\ &= 28 \text{ hours.} \\ \therefore \text{After 21 lessons, Claire will have} \\ &\text{completed 28 hours.} \end{aligned}$$

METHOD 2

$$\begin{aligned} S_n &= \frac{n}{2}[2a + (n-1)d] \\ a &= 30, d = 5, n = 21. \\ \therefore S_{21} &= \frac{21}{2}[2 \times 30 + (21-1) \times 5] \\ &= \frac{21}{2}(60 + 20 \times 5) \\ &= 1680 \text{ min} \\ &= 28 \text{ hours.} \\ \therefore \text{After 21 lessons, Claire will have} \\ &\text{completed 28 hours.} \end{aligned}$$

(iii) 50 hours = 50×60 minutes
= 3000 minutes.
 $a = 30$, $d = 5$, $S_n = 3000$, $n = ?$
Using $S_n = \frac{n}{2}[2a + (n-1)d]$

$$\begin{aligned} 3000 &= \frac{n}{2}[60 + (n-1) \times 5] \\ 6000 &= n(60 + 5n - 5) \\ 6000 &= n(55 + 5n) \\ 6000 &= 55n + 5n^2 \\ \therefore 5n^2 + 55n - 6000 &= 0 \\ n^2 + 11n - 1200 &= 0. \end{aligned}$$

Using the quadratic formula,

$$\begin{aligned} n &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-11 \pm \sqrt{11^2 - 4 \times 1 \times (-1200)}}{2 \times 1} \\ &= \frac{-11 \pm \sqrt{4921}}{2} \\ n &= 29.574 \ 919 \ 81 \dots \text{ or } \\ n &= -40.574 \ 919 \ 81 \dots \end{aligned}$$

(Ignore the negative answer as $n > 0$.)
∴ More than 29 lessons needed as $n > 29$.
ie. Clare will have completed 50 hours of lessons during her 30th lesson.

Q11

2005

Anne and Kay are employed by an accounting firm.

Anne accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by \$2500.

Kay accepts employment with an initial annual salary of \$50 000. In each of the following years her annual salary is increased by 4%.

- What is Anne's annual salary in her thirteenth year? 2
- What is Kay's annual salary in her thirteenth year? 2
- By what amount does the total amount paid to Kay in her first twenty years exceed that paid to Anne in her first twenty years? 3

(i) Anne's salary: 50 000, 52 500, 55 000, ...
∴ An arithmetic sequence with $a = 50\ 000$
 $d = 2500$.

$$\begin{aligned} T_n &= a + (n-1)d \\ \therefore T_{13} &= 50\ 000 + (13-1) \times 2500 \\ &= 50\ 000 + 12 \times 2500 \\ &= 80\ 000. \end{aligned}$$

∴ Anne's salary in her 13th year will be \$80 000.

(ii) Kay's salary: 50 000, $50\ 000 \times (1.04)$,
 $50\ 000 \times (1.04)^2$, ...
∴ A geometric sequence with $a = 50\ 000$
 $r = 1.04$.

$$\begin{aligned} T_n &= ar^{n-1} \\ T_{13} &= 50\ 000(1.04)^{13-1} \\ &= 80\ 051.610 \ 93 \dots \end{aligned}$$

∴ Kay's salary in her 13th year will be \$80 051.61 (nearest cent).

(iii) Kay's total salary in her first 20 years:

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right)$$

$$S_{20} = 50\ 000 \left[\frac{(1.04)^{20} - 1}{1.04 - 1} \right]$$

$$S_{20} = 1\ 488\ 903.929 \dots$$

∴ Kay's total salary in her first 20 years will be \$1 488 903.93.

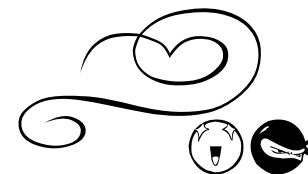
Anne's total salary in her first 20 years:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\begin{aligned} S_{20} &= \frac{20}{2}[2 \times 50\ 000 + (20-1) \times 2500] \\ &= 1\ 475\ 000. \end{aligned}$$

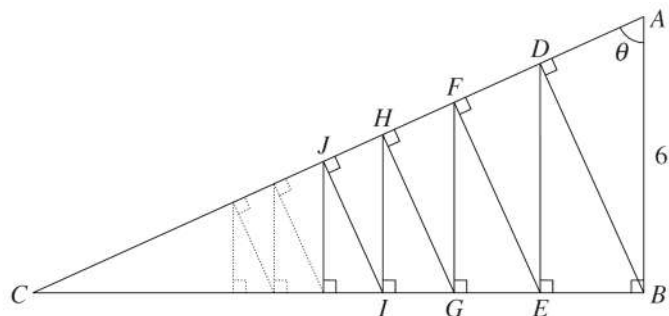
∴ Anne's total salary in her 20 years will be \$1 475 000.

∴ Kay's total salary exceeds Anne's total salary by \$1 488 903.93 - \$1 475 000 = \$13 903.93.



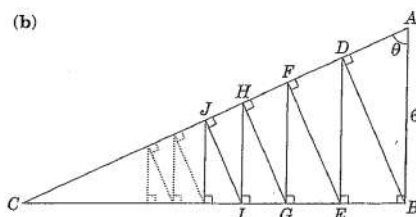
Q12

2005



The triangle ABC has a right angle at B , $\angle BAC = \theta$ and $AB = 6$. The line BD is drawn perpendicular to AC . The line DE is then drawn perpendicular to BC . This process continues indefinitely as shown in the diagram.

- (i) Find the length of the interval BD , and hence show that the length of the interval EF is $6 \sin^3 \theta$. 2
- (ii) Show that the limiting sum $BD + EF + GH + \dots$ is given by $6 \sec \theta \tan \theta$. 3



(i) $AB = 6$, $\angle BAC = \theta$ ($= \angle DAB$)

$$\text{In } \triangle ABD: \sin \theta = \frac{BD}{6}$$

$$\therefore BD = 6 \sin \theta.$$

$$\angle DAB + \angle ABD = 90^\circ \text{ (from } \triangle ABD)$$

$$\angle ABD + \angle DBE = 90^\circ \text{ (from } \triangle ABC)$$

$$\therefore \angle DBE = \theta.$$

$$\text{In } \triangle DEB: \sin \theta = \frac{DE}{DB}$$

$$\text{ie. } \sin \theta = \frac{DE}{6 \sin \theta}$$

$$\therefore DE = 6 \sin^2 \theta.$$

Similarly, in $\triangle EFD$:

$$\sin \theta = \frac{EF}{DE}$$

$$\text{ie. } \sin \theta = \frac{EF}{6 \sin^2 \theta}$$

$$\therefore EF = 6 \sin^3 \theta.$$

(ii) Note: $FG = 6 \sin^4 \theta$

$$GH = 6 \sin^5 \theta$$

$$\text{So } BD + EF + GH + \dots = 6 \sin \theta + 6 \sin^3 \theta + 6 \sin^5 \theta + \dots$$

This is a geometric series with $a = 6 \sin \theta$
 $r = \sin^2 \theta$.

This series has a limiting sum since $0 < \sin^2 \theta < 1$ as $0^\circ < \theta < 90^\circ$.

$$\begin{aligned} S_\infty &= \frac{a}{1-r} \\ &= \frac{6 \sin \theta}{1 - \sin^2 \theta} \\ &= \frac{6 \sin \theta}{\cos^2 \theta} \\ &= 6 \times \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta} \\ &= 6 \sec \theta \tan \theta. \end{aligned}$$

Q13

2006

On the first day of the harvest, an orchard produces 560 kg of fruit. On the next day, the orchard produces 543 kg, and the amount produced continues to decrease by the same amount each day.

- (i) How much fruit is produced on the fourteenth day of the harvest? 2
- (ii) What is the total amount of fruit that is produced in the first 14 days of the harvest? 1
- (iii) On what day does the daily production first fall below 60 kg? 2

(c) (i) 560, 543, 526,

Testing for an AP

$$\therefore T_2 - T_1 = T_3 - T_2$$

$$543 - 560 = 526 - 543 = -17 = d \therefore AP$$

$$T_n = a + (n-1)d$$

$$T_{14} = 560 + (14-1) \times -17$$

$$= 560 + 13 \times -17$$

$$= 339 \text{ kg}$$

(c) (iii)

$$T_n < 60$$

$$560 + (n-1) \times -17 < 60$$

$$560 - 17n + 17 < 60$$

$$-17n < -517$$

$$n > 30 \frac{7}{17}$$

$$n = 31 \therefore \text{on the 31}^{\text{th}} \text{ day}$$

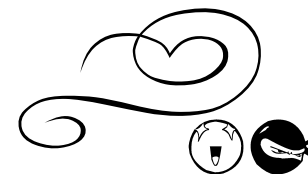
(ii)

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{14} = \frac{14}{2} [2 \times 560 + (14-1) \times -17]$$

$$= 7 [1120 + 13 \times -17]$$

$$= 6293 \text{ kg}$$



Q14

2006

Evaluate $\sum_{r=2}^4 \frac{1}{r}$.

$$\sum_{r=2}^4 \frac{1}{r} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = 1\frac{1}{12}$$

Q15

2008

Find the sum of the first 21 terms of the arithmetic series $3 + 7 + 11 + \dots$.

Q1f $S_n = \frac{n}{2}(2a + (n-1)d)$, where $a = 3$, $d = 4$ and $n = 21$.

$$S_{21} = \frac{21}{2}(2(3) + 20(4)) = 903$$

1

Q16

2007

Heather decides to swim every day to improve her fitness level.

On the first day she swims 750 metres, and on each day after that she swims 100 metres more than the previous day. That is, she swims 850 metres on the second day, 950 metres on the third day and so on.

- (i) Write down a formula for the distance she swims on the n th day. 1
- (ii) How far does she swim on the 10th day? 1
- (iii) What is the total distance she swims in the first 10 days? 1
- (iv) After how many days does the total distance she has swum equal the width of the English Channel, a distance of 34 kilometres? 2

$$(b)(i) T_n = a + (n-1)d$$

$$= 750 + (n-1) 100$$

$$= 650 + 100n$$

$$(ii) T_{10} = 650 + 100 \cdot 10 = 1650 \text{ m}$$

$$(iii) S_n = \frac{n}{2}(a + l)$$

$$S_{10} = \frac{10}{2}(750 + 1650) = 12000 \text{ m}$$

$$(iv) S_n = \frac{n}{2}[2a + (n-1)d]$$

$$34000 = \frac{n}{2}[2 \cdot 750 + (n-1) 100]$$

$$68000 = n[1500 + 100n - 100]$$

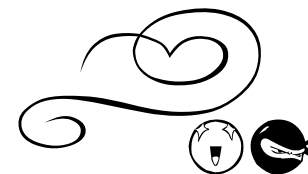
$$= n[1400 + 100n]$$

$$= 1400n + 100n^2$$

$$0 = n^2 - 14n - 680$$

$$(n+34)(n-20) = 0$$

$$n = 20$$



Q17

2008

The zoom function in a software package multiplies the dimensions of an image by 1.2. In an image, the height of a building is 50 mm. After the zoom function is applied once, the height of the building in the image is 60 mm. After a second application, its height is 72 mm.

- Calculate the height of the building in the image after the zoom function has been applied eight times. Give your answer to the nearest mm.
- The height of the building in the image is required to be more than 400 mm. Starting from the original image, what is the least number of times the zoom function must be applied?

Q4bi $50 \times 1.2^8 = 215 \text{ mm}$

Q4bii $50 \times 1.2^n > 400, 1.2^n > 8, n > \frac{\log 8}{\log 1.2} = 11.4.$

$\therefore$ 12 times at least.

Q18

2008

Consider the geometric series

$$5 + 10x + 20x^2 + 40x^3 + \dots$$

- For what values of x does this series have a limiting sum? 2
- The limiting sum of this series is 100. Find the value of x . 2

Q5bi $5 + 10x + 20x^2 + 40x^3 + \dots = 5(1 + (2x) + (2x)^2 + (2x)^3 + \dots).$

This series has a limiting sum if $-1 < 2x < 1. \therefore -\frac{1}{2} < x < \frac{1}{2},$

i.e. $|x| < \frac{1}{2}.$

Q5bii $a = 1, r = 2x, S_{\infty} = \frac{a}{1-r}.$

$\therefore \frac{1}{1-2x} = \frac{100}{5}, 1 = 20 - 40x, x = \frac{19}{40}.$


Q19

An arithmetic series has 21 terms. The first term is 3 and the last term is 53.

2009

Find the sum of the series.

Q3a Use $S_n = \frac{n}{2}(a + \ell)$, $S_{21} = \frac{21}{2}(3 + 53) = 588$.

Q20

 Evaluate $\sum_{k=1}^4 (-1)^k k^2$.

2009

Q2c $\sum_{k=1}^4 (-1)^k k^2 = (-1)^1 1^2 + (-1)^2 2^2 + (-1)^3 3^2 + (-1)^4 4^2$
 $= -1 + 4 - 9 + 16 = 10$.

2

Q21

A tree grows from ground level to a height of 1.2 metres in one year. In each

 subsequent year, it grows $\frac{9}{10}$ as much as it did in the previous year.

2009

Find the limiting height of the tree.

Q4a Limiting height is $S_\infty = \frac{a}{1-r} = \frac{1.2}{1-\frac{9}{10}} = 12$ m.

2

2

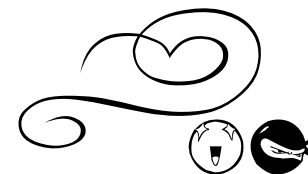
Q22

 Find the limiting sum of the geometric series $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} + \dots$.

2010

Q1f Geometric series: $a = 1$, $r = -\frac{1}{3}$, $S_\infty = \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$

2



Q23

2010

Susannah is training for a fun run by running every week for 26 weeks. She runs 1 km in the first week and each week after that she runs 750 m more than the previous week, until she reaches 10 km in a week. She then continues to run 10 km each week.

- How far does Susannah run in the 9th week?
- In which week does she first run 10 km?
- What is the total distance that Susannah runs in 26 weeks?

Q4ai Arithmetic sequence:

$$a = 1, d = 0.75, t_9 = 1 + 8 \times 0.75 = 7$$

$\therefore$ 7 km in the 9th week

Q4aii $t_n = 1 + (n-1)0.75 = 10, n = 13, \therefore$ the 13th week

Q4aiii $S_{13} = \frac{13}{2}(2(1) + (13-1)0.75) = 71.5$

Total distance = $71.5 + 13 \times 10 = 201.5$ km

Q24

2011

A skyscraper of 110 floors is to be built. The first floor to be built will cost \$3 million. The cost of building each subsequent floor will be \$0.5 million more than the floor immediately below.

- What will be the cost of building the 25th floor?
- What will be the cost of building all 110 floors of the skyscraper?

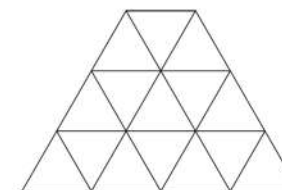
Q3ai $t_1 = 3, d = 0.5, t_{25} = 3 + 24 \times 0.5 = 15$, \$15 million

Q3aii $S_{110} = \frac{110}{2}(2 \times 3 + 109 \times 0.5) = 3327.5$, \$3327.5 million

Q25

2012

Jay is making a pattern using triangular tiles. The pattern has 3 tiles in the first row, 5 tiles in the second row, and each successive row has 2 more tiles than the previous row.



Row 1

Row 2

Row 3

- How many tiles would Jay use in row 20? 2
- How many tiles would Jay use altogether to make the first 20 rows? 1
- Jay has only 200 tiles. 2

How many complete rows of the pattern can Jay make?

Q12ci Arithmetic sequence: $a = 3, d = 2$

$$t_{20} = 3 + (20-1) \times 2 = 41$$

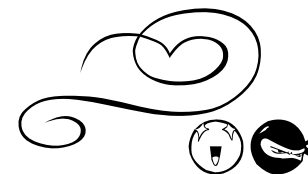
Q12cii $S_{20} = \frac{20}{2}(3 + 41) = 440$

Q12ciii $S_n = \frac{n}{2}(2 \times 3 + (n-1) \times 2) \leq 200$

$$n(6 + 2n - 2) \leq 400, 2n^2 + 4n \leq 400, n^2 + 2n + 1 \leq 201$$

$$(n+1)^2 \leq 201, n+1 \leq \sqrt{201}, n \leq \sqrt{201} - 1$$

$$\therefore n = 13$$


Q26
2012

Rectangles of the same height are cut from a strip and arranged in a row. The first rectangle has width 10 cm. The width of each subsequent rectangle is 96% of the width of the previous rectangle.



10 cm

NOT TO
SCALE

- (i) Find the length of the strip required to make the first ten rectangles.
- (ii) Explain why a strip of length 3 m is sufficient to make any number of rectangles.

2

1

Q15ai Geometric series: $a = 10$, $r = 96\% = 0.96$

$$S_{10} = \frac{10(1 - 0.96^{10})}{1 - 0.96} \approx 83.8 \text{ cm}$$

The length required is 83.8 cm approximately.

Q15aii $S_{\infty} = \frac{10}{1 - 0.96} = 250 \text{ cm or } 2.5 \text{ m}$

$\therefore$ a 3-m strip is sufficient to make any number of rectangles.

Q27
2013

Kim and Alex start jobs at the beginning of the same year. Kim's annual salary in the first year is \$30 000, and increases by 5% at the beginning of each subsequent year. Alex's annual salary in the first year is \$33 000, and increases by \$1500 at the beginning of each subsequent year.

- (i) Show that in the 10th year Kim's annual salary is higher than Alex's annual salary.
- (ii) In the first 10 years how much, in total, does Kim earn?
- (iii) Every year, Alex saves $\frac{1}{3}$ of her annual salary. How many years does it take her to save \$87 500?

2

2

3

Q12ci Kim: $t_1 = 30000$, $r = 1 + 0.05 = 1.05$

$$t_{10} = 30000 \times 1.05^9 \approx 46539.85 \text{ dollars}$$

Alex: $t_1 = 33000$, $d = 1500$

$$t_{10} = 33000 + 9 \times 1500 = 46500.00 \text{ dollars}$$

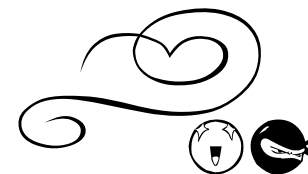
Q12cii Kim: $S_{10} = \frac{30000(1.05^{10} - 1)}{1.05 - 1} \approx 377336.77 \text{ dollars}$

Q12ciii Alex's n year savings:

$$\frac{1}{3}S_n = 87500, \frac{1}{3} \times \frac{n}{2}(2 \times 33000 + (n-1)1500) = 87500 \text{ and } n > 0$$

$$n^2 + 43n - 350 = 0, (n-7)(n+50) = 0, \therefore n = 7 \text{ years.}$$





Q28

2014

Evaluate the arithmetic series $2 + 5 + 8 + 11 + \dots + 1094$.

Q12a $a = 2$, $d = 3$, $t_n = 1094$, $\therefore 2 + (n-1)3 = 1094$, $n = 365$

$$S_{365} = \frac{365}{2}(2 + 1094) = 200020$$

Q29

2015

Find the limiting sum of the geometric series $1 - \frac{1}{4} + \frac{1}{16} - \frac{1}{64} + \dots$.

$$S_{\infty} = \frac{1}{1 - \left(-\frac{1}{4}\right)} = \frac{4}{5}$$

Q30

2016

By summing the geometric series $1 + x + x^2 + x^3 + x^4$, or otherwise, find $\lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1}$.

Q14d $1 + x + x^2 + x^3 + x^4 = \frac{1(x^5 - 1)}{x - 1}$

$$\therefore \lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1} = \lim_{x \rightarrow 1} (1 + x + x^2 + x^3 + x^4) = 5$$

Q31

2017

In an arithmetic series, the fifth term is 200 and the sum of the first four terms is 1200. Find the value of the tenth term.

$$T_5 = a + 4d = 200$$

$$S_5 = a + a + d + a + 2d + a + 3d = 1200$$

$$a + 4d = 200 \quad \text{--- ①}$$

$$4a + 6d = 1200 \quad \text{--- ②}$$

$$4a + 16d = 800 \quad \text{--- ①} \times 4$$

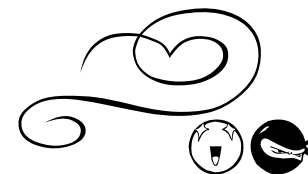
Subtracting $10d = -400$
 $d = -40$

$$a + -160 = 200$$

$$a = 360$$

$$T_{10} = 360 + 9 \times -40$$

$$= 0$$



Q32

2018

In an arithmetic series, the third term is 8 and the twentieth term is 59.

- (i) Find the common difference.
- (ii) Find the 50th term.

Question 11 (d) (i)

Sample answer:

$$a + 2d = 8 \quad \textcircled{1}$$

$$a + 19d = 59 \quad \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}: \frac{17d}{17} = \frac{51}{17}$$

$$d = 3$$

Question 11 (d) (ii)

Sample answer:

$$d = 3, \text{ so } a + 2(3) = 8$$

$$a + 6 = 8$$

$$a = 2$$

$$a + 49d = 2 + 49 \times 3$$

$$= 2 + 147$$

$$= 149$$

Q33

2019

What is the limiting sum of the following geometric series?

$$2000 - 1200 + 720 - 432 \dots$$

$$a = 2000, r = -0.6, S_{\infty} = \frac{2000}{1 - (-0.6)} = 1250$$

Q34

2019

In an arithmetic series, the fourth term is 6 and the sum of the first 16 terms is 120. Find the common difference.

Q12b Solve the simultaneous equations $t_4 = a + 3d = 6$

$$\text{and } S_{16} = 8(2a + 15d) = 120, \therefore d = \frac{1}{3}$$